

PROJECT REPORT

SMART INVENTORY MANAGEMENT SYSTEM

Name: Ayush

Reg. No.: 25BAI10877

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1. Abstract

Smart Inventory Management System is a command-line-based Java application designed for managing products, inventory, customer orders, and inventory transactions. The system offers product addition, viewing, searching, updating, and deletion operations, as well as stock-in/stock-out functionality. It also includes low stock detection, inventory value calculation, order placing, checking availability, and recording transactions. Simple and portable data persistence, without the need of an external database, using CSV files. The project illustrates the use of Java objects and classes, collections, exception handling, file I/O, input validation, and synchronized operations.

2. Introduction

Managing inventory is a crucial task for businesses, as it involves keeping track of the products that are available and in stock. The more products and transactions, the more difficult manual inventory records become. This project adopts a straightforward solution using Java, which is fully terminal driven. The application stores product information in CSV files and has separate menus for product management, stock management, order management, and reports.

3. Objectives

- Develop a command-line inventory management application in Java.
- Add, view, search, update, and delete product records.
- Provide stock-in and stock-out operations.
- Detect products whose stock is below the defined low-stock level.
- Calculate the total value of the available inventory.
- Place customer orders after checking product availability.
- Maintain transaction records and store data using CSV files.
- Apply input validation and exception handling for invalid operations.

4. Methodology

The application's design is based on the object-oriented principles. Product, Order and Transaction are implemented as Java classes. Inventory class manages products and stock operations, InventoryFile class is used for reading and writing CSV file. User input is provided to the command-line menu and the menu puts the user's choice into the inventory operations with the help of the Main function. In-memory data is stored in collections like ArrayList, Vector, and Stack, as needed. Validation is done prior to significant operations like removal from the stock and placing orders. Operations relating to critical stocks are synchronized.

5. Project Modules

Main: Provides the command-line menu and accepts user input.

Product: Stores product ID, name, category, price, quantity, and low-stock level.

Inventory: Manages products, stock operations, searching, reports, and order stock deduction.

Order: Stores customer order details and ordered quantities.

Transaction: Stores stock movement and order transaction information.

InventoryFile: Reads and writes product, transaction, and order data in CSV format.

6. Technologies Used

- Java (JDK 17 or later)
- Object-Oriented Programming
- Java Collections Framework
- Exception Handling and Input Validation
- Java File I/O and NIO
- CSV file storage
- Command-line interface

7. Implementation and Working

When the program starts, it loads product information from the CSV storage file if it's there at the beginning of the program. The main menu offers four functional areas: Product Management, Stock Management, Order Management, and Reports. Records can be added and edited in Product Management. Stock management is used to update quantities and to record stock transactions. Order management ensures that there is sufficient quantity before decreasing stock and keeps track of the order. Reports show complete inventory, low stock, and total inventory value. Whenever data is changed, the relevant CSV files are updated.

8. Results and Discussion

The system used has been implemented and is able to help with the project's core inventory functions. Maintenance and stock can be manipulated using the command line, and an insufficient stock count is rejected rather than accepted for an invalid operation. When you place an order, inventory is deducted from available stocks, and when you conduct inventory, it will give you great information on inventory value and current stocks available. The project is simple to implement, as the storage is done using CSV.

9. Testing

There is a basic validation test class in the tests folder. The tests are done to ensure product creation and search, stock-in, stock-out, and rejection of an operation when stock is less than requested. The project was compiled with the Java compiler javac, "All," and the validation test was successfully finished with the message "All validation tests passed."

10. Future Improvements

- Add a graphical user interface for easier interaction.
- Add database support for larger-scale persistent storage.
- Add user login and role-based access control.
- Add advanced sales and inventory analytics.
- Add automatic backup and restore of inventory data.
- Improve reporting with exportable summaries and charts.

11. Conclusion

The Smart Inventory Management System is a practical application and example of how Java can be used to create a command-line application. This project integrates object-oriented programming, collections,

exception handling, validation, synchronization, and file handling into a single application. The use of CSV files instead of an external database makes it easier and simpler to set up and run the system without compromising on the necessary function to manage inventory.

12. Project Structure

SmartInventoryManagementSystem/

```
|
|— src/
|   |— Main.java
|   |— Product.java
|   |— Inventory.java
|   |— Order.java
|   |— Transaction.java
|   |— InventoryFile.java
|
|— tests/
|   |— InventoryTest.java
|
|— README.md
|— .gitignore
```